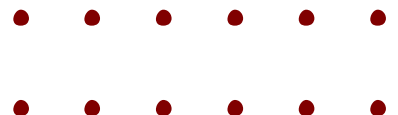




AI-Powered Intelligent Job Search and Career System
Data Science and AI Project
Milestone 3 Report
Group 4

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1. Introduction

1.1 Brief Introduction of the project

The AI-Powered Intelligent Job Search and Career System connect three core modules: Resume Parsing, Career Recommendation, and Job Discovery and Job Matching. The Resume Parsing module converts an uploaded resume into a structured candidate profile. The Career Recommendation module maps that profile to suitable occupational roles. The selected roles then guide the Job Discovery and Job Matching module when retrieving and ranking job postings. This shared-profile architecture was established in Milestone 1, and the required datasets were prepared and validated in Milestone 2.

Resume Parsing is the entry point of the entire pipeline. The goal of this module is accurate, structured, privacy-respecting extraction from resumes that arrive in inconsistent formats. A defining constraint of the working dataset (‘hadikp/resume-data-pdf’) is that it is 100% single-page, image-only PDFs with no text layer. This makes the module a document-understanding / vision task, not a plain-text NER task. The model architecture is therefore chosen because it is multimodal, instruction-following, and able to return schema-constrained JSON properties that map directly onto "turn any resume into one clean candidate profile

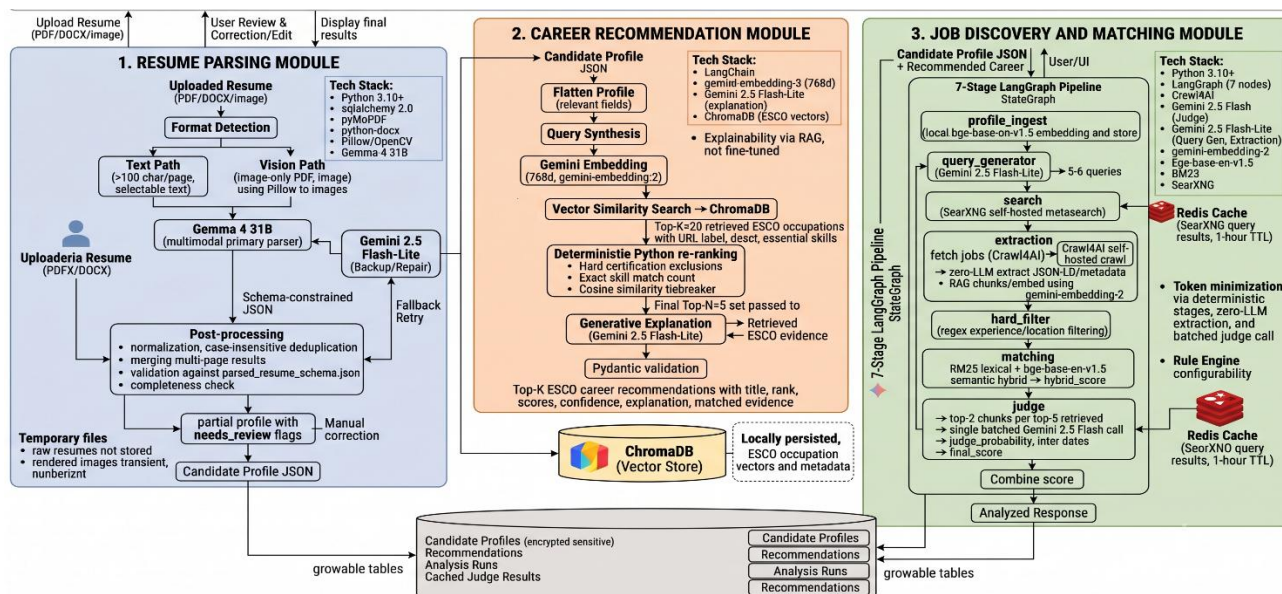
For Career Recommendation, Milestone 2 prepared the ESCO v1.2.1 English classification as a retrieval knowledge base containing 3,039 occupations, 13,939 cleaned skills, and 126,051 valid occupation-skill relationships. Every prepared occupation has linked skills, and the occupation set covers all ten ISCO major groups. ESCO is used as a general occupational taxonomy rather than as a supervised training dataset.

The Job Discovery and Matching module evaluates candidate-job fit and produces a scored, explained fit report including an overall interview probability score, a breakdown by category, matched and gap skills, and a recruiter-style written recommendation. The module receives the Candidate Profile and the recommended career/occupation as inputs. It matches against a local job database first, and falls back to web discovery via Crawl4AI when no suitable jobs exist locally.

1.2 Objectives of Milestone 3

- Finalise the architecture of all the modules and projects and define how it interacts with each other.
- Specify retrieval, deterministic re-ranking, prompt engineering, structured output, error handling, storage, and API contracts.
- Define the complete model architecture for the candidate-job fit pipeline.

2. Overall System Architecture



2.1 Major modules and interactions

Module	Receives	Produces	Interaction with next stage
Resume Parsing	Uploaded resume	Structured candidate profile	Supplies the common Candidate Profile JSON.
Career Recommendation	Candidate Profile JSON	Top-K ESCO career recommendations	Passes recommended roles and explanations to Job Discovery and Matching.
Job Discovery and Matching	Candidate profile + recommended roles	Ranked jobs and match explanations	Returns the fit score, category breakdown, matched and gap skills, and recruiter-style recommendation to the user interface.

2.2 Tech Stack

Module	Layer	Stacks	Purpose
All Module	Programming language	Python 3.10+	Dataset handling, indexing, API logic, and evaluation, Rule engine
All Module	Application database	PostgreSQL	Stores candidate profiles, recommendations, confidence values, and timestamps.
All Module	ORM	SQLAlchemy 2.0	Async-compatible database sessions.
All Module	Cache	Redis	SearXNG query result cache (1 hour).
Resume Parsing	Primary parser	Gemma 4 31B (multimodal)	Pre-trained, open-weight, zero-shot resume extraction.
Resume Parsing	Backup / repair	Gemini 2.5 Flash-Lite	Pre-trained hosted API used only on fallback.

Resume Parsing	Document processing	PyMuPDF, python-docx, Pillow/OpenCV	Routing, PDF rendering, text extraction, and optional image enhancement.
Career Recommendation	Embedding model	gemini-embedding-2 with 768 output dimensions	Creates candidate and occupation vectors in a shared semantic space.
Career Recommendation	Vector database	ChromaDB persistent collection	Stores ESCO occupation vectors and metadata.
Career Recommendation	Explanation model	Gemini 2.5 Flash-Lite	Generates grounded explanations from retrieved ESCO evidence.
Career Recommendation	Orchestration	LangChain components	Connects preprocessing, retrieval, prompts, parsing, and validation.
Job Discovery and Matching	Orchestration	LangGraph (StateGraph), 7 nodes	Wires the pipeline with shared state and per-node error short-circuiting.
Job Discovery and Matching	Search	SearXNG (self-hosted)	Metasearch across multiple engines without a paid search API.
Job Discovery and Matching	Crawling	Crawl4AI	Renders JS-heavy job boards; produces pruned, ad/nav-stripped markdown.
Job Discovery and Matching	LLM	Gemini 2.5 Flash-Lite/ Gemini 2.5 Flash	Generates search queries from the candidate profile., Single batched call scoring the top 5 shortlisted jobs.
Job Discovery and Matching	Embeddings	BAAI/bge-base-en-v1.5	Local bi-encoder for candidate/job similarity; no API call.
Job Discovery and Matching	Lexical ranking	BM25 (rank_bm25)	Exact keyword/skill matching component of the hybrid score.

3. End-to-End Workflow

The system follows a three-module pipeline. First, the Resume Parsing module processes the uploaded resume and produces a validated Candidate Profile. Next, the Career Recommendation module uses the Candidate Profile to retrieve and rank suitable ESCO occupations and generate grounded career recommendations. Finally, the Job Discovery and Matching module receives the Candidate Profile and a recommended job, extracts the structured Job Profile, calculates deterministic rule-based scores, applies the hard eligibility gate, and uses Gemini for recruiter-style fit evaluation. The final combined score, strengths, gaps, recommendation, and gate status are stored in PostgreSQL and returned to the frontend.

3.1 Resume Parsing Module

It is the only module that touches the raw uploaded file, and it is the producer of the shared Candidate Profile object that the other two modules consume.

Complete workflow (upload → profile)

Upload: PDF / DOCX / image → format detection → text-path or vision-path routing → preprocessing → Gemma 4 extraction with schema-constrained JSON → validation and completeness check → multi-page merge, normalisation, and skill deduplication → PostgreSQL persistence → deletion of temporary rendered images.

For DOCX files, python-docx extracts text. For PDFs, PyMuPDF checks whether each page has at least 100 extractable characters. Text PDFs use the text path; image-only PDFs use the vision path. Image files and rendered PDF pages are sent one page at a time to Gemma 4.

When Gemma 4 output fails jsonschema validation or the completeness check, Gemini 2.5 Flash-Lite performs one fallback retry using the same schema. If both models fail, the module emits a partial profile with needs_review flags so the user can correct it manually.

The validated Candidate Profile is shown for user review and correction before it is released to Career Recommendation and Job Discovery and Matching. The system never auto-fabricates missing sections; absent sections are shown as empty and editable.

3.2 Career Recommendation Module

- The module accepts a structured dictionary representing the user's professional profile.
- The module flattens relevant profile fields into a comprehensive semantic query string.
- The query string is vectorized using the Gemini embedding model.
- ChromaDB performs a vector similarity search against the pre-indexed ESCO taxonomy.
- A deterministic Python re-ranking step filters out occupations with unmet hard requirements, then orders the remainder by exact skill-match count against ESCO essential skills, using cosine similarity as a tiebreaker.
- The final re-ranked occupations and their supporting ESCO evidence are passed to the Gemini model, which generates grounded explanations and potential skill-gap observations in the required structured format.

3.3 Job Discovery and Matching Module

The module receives the Candidate Profile (including the recommended career/occupation from Career Recommendation) and returns a ranked shortlist of the five best-fit job openings, each with an interview-probability score and a recruiter-style rationale. It runs as a seven-stage LangGraph pipeline, sharing one state object across stages. A failed stage sets an error field that downstream stages check before executing, so a failure short-circuits the run instead of raising.

1. **Profile Ingest:** The candidate profile is embedded using the local bge-base-en-v1.5 bi-encoder and stored.
2. **Query Generator:** Gemini 2.5 Flash-Lite generates 5–6 diverse search queries (role variants, skill-based, domain-specific) from the candidate profile and recommended career/occupation.
3. **Search Module:** Each query runs against the self-hosted SearXNG metasearch engine; results are cached in Redis for 1 hour.
4. **Extraction Module:** The global jobs table is checked first by job_key (a stable hash of the normalized URL); on a miss, Crawl4AI crawls the posting, structured fields are extracted from JSON-LD/page metadata, and the posting is split into RAG chunks and embedded.
5. **Hard Filter:** Regex-based experience-range and location/remote filtering, so vague postings pass through rather than being wrongly excluded.

6. **Matching Module:** Hybrid ranking combines BM25 (exact skill match) with bge-base-en-v1.5 cosine similarity (semantic match): $\text{hybrid_score} = (0.4 \times \text{BM25} + 0.3 \times \text{embedding}) / 0.7$. The top 15 jobs are kept.
7. **Judge Module:** For the top 5 shortlisted jobs, the top-2 most relevant RAG chunks per job are retrieved by cosine similarity to the candidate's embedding. A single batched Gemini 2.5 Flash call scores all 5 finalists at once and infers any missing posted/expiry dates not already found via JSON-LD or regex.

$$\text{final_score} = 0.8 \times \text{hybrid_score} + 0.2 \times \text{judge_probability}$$

Only two Gemini calls happen per run regardless of how many jobs are discovered — every other stage (embedding, search, crawling, extraction, filtering, ranking) is deterministic and model-free by design, since token cost was a defining engineering constraint on this module.

Global job store: the jobs table is a persistent system of record shared across all candidates, not recreated per session. A posting is crawled and embedded once, then reused by every subsequent search that matches it, until it goes stale (configurable TTL) or its listed expiry date passes.

3.4 Inputs and Outputs of Each Module

Module	Input	Output
Resume Parsing	Uploaded resume file (PDF / DOCX / image).	A single validated Candidate Profile JSON containing contact, skills, education, experience, projects, certifications, and job_titles.
Career Recommendation	Structured Candidate Profile dictionary containing lists of skills, experiences, and a professional summary	A list of dictionaries containing recommended ESCO occupation codes, official titles, match score from 0 to 100 and a confidence band. The match score is a deterministic decision-support score and not a statistical probability.
Job Discovery and Matching	Candidate Profile JSON + recommended career/occupation	Ranked shortlist of 5 jobs, each with final_score, hybrid_score, judge_probability, and a recruiter-style rationale

3.5 Error Handling and Fallback Mechanisms

Module	Failure condition	Handling
Career Recommendation	Missing or invalid profile fields	Return a validation response and request profile correction before inference.
Career Recommendation	Embedding API timeout	Retry with bounded exponential backoff; do not create a partial recommendation.
Career Recommendation	Vector store unavailable	Return a service-unavailable status and preserve the request for later retry.
Career Recommendation	Weak retrieval scores	Broaden Top-K retrieval and return a low-confidence flag rather than forcing a role.
Career Recommendation	Invalid or non-schema LLM output	Run structured-output validation and one repair attempt; otherwise use retrieval-only results.

Career Recommendation	Duplicate recommendations	Deduplicate by ESCO occupation URI during post-processing.
Resume Parsing	Corrupt / unreadable file	Reject with user-facing message; no model call.
Resume Parsing	Image-only PDF misrouted as text	If fewer than 100 extractable characters per page are detected, automatically route to the vision path.
Resume Parsing	Incomplete text extraction	Use a vision-based retry.
Resume Parsing	Model returns invalid JSON	Repair via Gemini 2.5 Flash-Lite retry using the same schema.
Resume Parsing	Low field completeness	Use Gemini 2.5 Flash-Lite retry; otherwise emit a partial profile and flag fields for review.
Resume Parsing	Multi-page resume	Parse each page separately, then merge and deduplicate fields.
Resume Parsing	Both models fail	Persist a partial profile with needs_review flags so the user can edit it manually.
Job Discovery and matching	SearXNG returns no results for all generated queries	Return an empty shortlist with a "no matches found" status rather than failing the request.
Job Discovery and matching	Crawl4AI fails to render a posting	Skip that URL and continue with remaining candidates; do not fail the whole run for one bad page.
Job Discovery and matching	Gemini API rate limit or quota	Return an HTTP 503 response; do not silently retry in a loop.
Job Discovery and matching	PostgreSQL write fails	Log the error and still return the ranked shortlist to the user; a database write failure must not block the response.

3.6 Storage and retrieval

PostgreSQL stores the validated Candidate Profile and the outputs generated by the downstream modules. Retained sensitive fields, including company, institution, location, and links, are protected using field-level encryption and are decrypted only for the authenticated owner. Rendered resume-page images are temporary and are deleted immediately after parsing. The validated profile may be embedded once and cached using a content hash for reuse by downstream modules.

The ESCO vector index is persisted locally in ChromaDB and opened by the application at runtime for low-latency career-recommendation retrieval.

For Job Discovery and Matching, PostgreSQL stores each analysis run, including the Candidate Profile JSON, Job Profile JSON, rule score, judge score, final score, gated status, gate reason, creation timestamp, and a hash of the candidate and job JSON objects used for caching repeated analyses.

4. Model Architecture Selection

4.1 Module-wise model selection summary

Module	Model architecture	Pre-trained or custom-built
Resume Parsing	Primary parser: Gemma 4 31B (multimodal), a pre-trained, open-weight, zero-shot model. Backup / repair: Gemini 2.5 Flash-Lite, invoked only when Gemma 4 output fails schema validation or completeness checks. Text-layer probe: PyMuPDF. OCR: Tesseract for EDA only, not in the serving path. The pipeline splits PDFs into pages, renders one page to an image, sends one page image at a time with the schema prompt, and merges the per-page JSON.	Pre-trained models; zero-shot schema-constrained inference. No custom model is trained.
Career Recommendation	Embedding Model: gemini-embedding-2 via LangChain's GoogleGenerativeAIEmbeddings, configured to generate 768-dimensional dense vectors. It embeds candidate profiles and ESCO occupation documents into a shared semantic space for cosine-similarity retrieval. Generative Explanation Model: Gemini 2.5 Flash-Lite via LangChain's ChatGoogleGenerativeAI. A deterministic Python-based scoring function first re-ranks the retrieved occupations using semantic similarity, skill overlap, and title alignment. The Gemini model then generates grounded explanations and missing-skill insights in a structured JSON format; it does not independently calculate match scores or select unsupported occupations. Integration: The components are integrated through a LangChain execution chain, such as RunnableSequence, together with explicit Python retrieval, re-ranking, and validation functions. Candidate information is embedded, relevant ESCO occupations are retrieved from ChromaDB, deterministic re-ranking is applied, and the selected evidence is passed to the language model. The final structured output is validated using a Pydantic or JSON schema.	Pre-trained models; custom retrieval and scoring logic
Job Discovery and Matching	Query Generator: Gemini 2.5 Flash-Lite generates 5–6 search queries from the candidate profile. Search: SearXNG (self-hosted metasearch), cached in Redis. Extraction: zero-LLM, Crawl4AI crawls on a job-store miss, structured fields come from JSON-LD/page metadata, postings are chunked and embedded for RAG. Hard Filter: regex-based experience/location filtering. Matching: $\text{hybrid_score} = (0.4 \times \text{BM25} + 0.3 \times \text{bge-base-en-v1.5 embedding}) / 0.7$, keeps top 15. Judge: a single batched Gemini 2.5 Flash call scores the top 5 finalists using retrieved RAG chunks; $\text{final_score} = 0.8 \times \text{hybrid_score} + 0.2 \times \text{judge_probability}$.	Pre-trained Gemini models (query generation, judge) and a pre-trained local embedding model (bge-base-en-v1.5). Deterministic BM25 and regex logic. No fine-tuning is performed.

4.2 Resume Parsing Module

4.2.1 Components

- Ingestion & routing: Accept upload, detect format, and decide the text path or vision path.
- Preprocessing: Render PDF pages to images or extract text; enhance image quality only when required.
- Extraction engine: Gemma 4 multimodal inference with a schema-constrained prompt.
- Post-processing & validation: Normalise output, validate against `parsed_resume_schema.json`, and merge multi-page results.
- Fallback controller: On validation failure or low completeness, retry via Gemini Flash.
- Persistence: Write the validated profile to PostgreSQL and delete temporary rendered images.

4.2.2 Model size and complexity

Gemma 4 is available in two multimodal sizes, 26B and 31B. We use 31B as the primary parser: the extra capacity improves accuracy on the sparse, harder fields (projects, certifications), and because we send one page per request, model latency is not a concern for our workload. The 26B variant is kept as a lighter alternative.

4.2.3 Integration between models

Gemma 4 and Gemini 2.5 Flash-Lite never run together. Gemini 2.5 Flash-Lite is invoked only when Gemma 4's output fails schema validation or the completeness check, a strict fallback, not an ensemble. Both are bound to the same schema and same prompt contract, so downstream modules see identical output regardless of which produced it.

4.2.4 Why a Decoder-Only Vision-Language Architecture

The resume-parsing task has two hard requirements that shape the model choice: the page must be read as an image, since the source PDFs have no text layer, and the output is a nested JSON object whose size varies with the resume's length and content.

Encoder-only models are not a fit for this task: they label input tokens rather than generate new ones, so they cannot produce a variable-length, nested document as output. Gemma 4 is a decoder-only vision-language model, and its architecture matches both requirements directly: a vision encoder attends across the page's visual patches so layout and spatial structure are preserved, while the causal decoder writes the output JSON token by token, which is what allows the output to be constrained to the required schema.

Gemma 4 conditions on the page through the prompt prefix rather than through cross-attention between separate encoder and decoder stacks. In practice this produces the same effect as a traditional encoder-decoder architecture, but with a single set of model weights rather than two separately trained stacks simpler to deploy and reason about for this use case.

4.3 Career Recommendation Module

4.3.1 Components

Component	Input	Output
GoogleGenAIEmbeddings	Synthesized Python string containing candidate skills, summary, and experience details.	A dense mathematical array of floating-point numbers (1 x 768 dimensions).

ChromaDB Vector Store	Query vector (1 x 768).	List objects containing ESCO URI, official titles, descriptions, and similarity distance scores.
ChatGoogleGenerativeAI	LangChain SystemMessage and HumanMessage containing the composite profile text paired with the retrieved text blocks.	A strongly typed JSON string containing the structured match assessment.
Deterministic re-ranker	Retrieved occupations, semantic scores and normalized profile evidence	Occupations filtered by hard-requirement exclusions and ordered by exact skill-match count (cosine similarity as tiebreaker)
Pydantic/schema validator	Structured Gemini response and retrieved ESCO evidence	Validated recommendation objects containing only supported occupations and skills

4.3.2 Model size and complexity

This module works on RAG based architecture. No model training or fine-tuning is needed here.

Why retrieval-augmented generation over fine-tuning

1. Data requirements. Fine-tuning a model to map candidate profiles to ESCO occupations would require a labelled training set of (profile → correct occupation) pairs at meaningful volume per class. With 3,039 ESCO occupations, this would demand thousands of labelled candidate profiles distributed across every occupation to avoid class imbalance — data the project does not have and cannot ethically fabricate. RAG requires no labelled candidate-to-occupation pairs at all; it only needs the ESCO taxonomy itself, which is already curated and complete.

2. Ease of knowledge updates: ESCO is versioned and updated periodically (e.g., v1.1.1 → v1.2.1). A fine-tuned model would need to be retrained — or at minimum have its output layer reworked — every time the occupation taxonomy changes, incurring both compute cost and the risk of catastrophic forgetting on previously learned mappings. Under RAG, an ESCO update is handled by re-embedding and swapping the vector index, with zero change to the model weights and zero downtime.

3. Explainability: A fine-tuned model's occupation prediction is an opaque function of its trained weights — there is no direct way to show a user **why** a specific role was suggested beyond a confidence score. RAG keeps every recommendation traceable to specific retrieved ESCO records: the exact occupation URI, its essential skills, and the deterministic re-ranking logic, are all inspectable and can be shown to the user as evidence.

4. Lower maintenance: A fine-tuned model requires an ongoing MLOps loop: labelled data collection, retraining, evaluation, and redeployment each time drift is detected. The RAG pipeline's "training" surface is limited to the deterministic re-ranking rules and prompt design, both of which are plain Python/text artifacts that can be edited, tested, and deployed without a training run.

4.3.3 Integration between models

The same embedding model configuration must be used for both offline ESCO indexing and online candidate queries. Retrieved occupation evidence is passed to the explanation model, but the explanation model is not allowed to introduce occupations outside the retrieved candidate set.

4.4 Job Discovery and Matching

4.4.1 Components

Component	Input	Output
Query Generator (Gemini 2.5 Flash-Lite)	Candidate Profile + recommended career/occupation	5–6 diverse search query strings
Search Module (SearXNG)	Query strings	Candidate URLs, cached in Redis
Extraction Module (Crawl4AI + zero-LLM parsing)	Job posting URLs	Structured JobProfile fields (JSON-LD/metadata) + RAG chunks
Hard Filter (regex)	JobProfile list	Filtered JobProfile list
Matching Module (BM25 + bge-base-en-v1.5)	Filtered JobProfile list + candidate embedding	Top 15 jobs with hybrid_score
Judge Module (Gemini 2.5 Flash, batched)	Top 15 jobs + retrieved RAG chunks	Top 5 jobs with judge_probability and rationale

4.4.2 Model size and complexity

Gemini 2.5 Flash-Lite and Gemini 2.5 Flash are both hosted, so no GPU is required for the LLM stages. bge-base-en-v1.5 is a small, local bi-encoder (110M parameters) that runs on CPU with modest memory requirements, avoiding both GPU cost and per-call API billing for embeddings.

4.4.3 Integration between models

The pipeline is deliberately designed so that only two stages ever call an LLM, regardless of how many jobs are discovered or filtered: the Query Generator (Gemini 2.5 Flash-Lite, once per run) and the Judge Module (Gemini 2.5 Flash, once per run, batched across all 5 finalists). Every other stage like embedding, search, extraction, filtering, and hybrid ranking, is a deterministic function with no LLM involvement. This reduced a typical run's LLM call count from an estimated 45+ (one extraction call per discovered job, plus one judge call per shortlisted job, under a naive per-item design) down to exactly 2.

5. Justification of Model Choices

5.1 Resume Parsing

Gemma 4 is selected because the working dataset is image-only and therefore requires multimodal document understanding. The model reads full resume pages directly without a separate OCR stage, supports long context and schema-constrained JSON output, and can be used zero-shot without task-specific model training.

Training is unnecessary, not impossible. A strong instruction-following multimodal model already generalises to resume extraction zero-shot, so the system avoids the cost of building, labelling for, and maintaining a custom-trained model.

Gemini 2.5 Flash-Lite is used only as a backup or repair path. When the primary output fails validation, a second independent model is more likely to succeed than re-sampling the same model.

The alternatives considered were LayoutLM / LayoutLMv3, Donut, spaCy / BERT NER over OCR text, and regex or rule-based parsers. They were not selected because they require task-specific fine-tuning, field-level labels, a separate OCR stage, or brittle template-specific rules.

Target performance is $F1 \geq 0.75$ on the 52-resume held-out test set, reported per field and as an overall score.

5.2 Career Recommendation Model

Choice	Reason selected	Alternative considered	Reason not selected as primary
Semantic retrieval over ESCO	Supports thousands of occupations without needing labelled resumes for every class.	3,039-class supervised classifier	Insufficient labelled candidate profiles per occupation and poor flexibility for unseen profiles.
Gemini embedding model	Produces semantic vectors suitable for retrieval and supports a configurable 768-dimensional output.	TF-IDF or BM25 only	Lexical methods miss paraphrases and related skills; they may still be retained as an evaluation baseline.
ChromaDB	Simple persistent local vector store suitable for the project scale.	FAISS-only index or managed cloud vector database	FAISS needs extra metadata persistence; managed services add cost and operational complexity.
Deterministic re-ranking	Makes role order more explainable and reduces dependence on one similarity value.	LLM-only role ranking	LLM-only ranking is less reproducible and harder to evaluate.
Grounded explanation LLM	Produces readable recommendations while using retrieved ESCO evidence.	Template-only explanations	Templates are deterministic but may be less personalized; they remain a fallback.

5.3 Job Discovery and Matching Model

The guiding design principle is token minimization: never call an LLM where a deterministic method achieves comparable results, and where an LLM call is unavoidable, batch it across every item that needs it rather than issuing one call per item.

- **Why SearXNG instead of a paid search API or LLM-driven scraping:** SearXNG is a self-hosted metasearch engine, avoiding both the recurring cost of a paid search API and the unreliability of asking an LLM to construct scrape targets directly. It also gives the pipeline results from multiple underlying search engines in one call.
- **Why zero-LLM extraction:** most job boards already embed structured data via schema.org JobPosting JSON-LD, or expose it in consistent page metadata. Parsing this directly is more reliable and faster than an LLM extraction call, and costs nothing per job.

- **Why a hybrid BM25 + embedding score instead of an LLM-only ranking:** BM25 captures exact skill/keyword matches that embeddings alone can miss (e.g., an exact tool or certification name), while the embedding component captures semantic similarity for differently-worded but equivalent skills. Combining both, entirely without an LLM call, lets the system rank the full candidate pool (potentially hundreds of jobs) at negligible cost before any LLM is involved.
- **Why two Gemini tiers, one query-generation call and one batched judge call:** query generation is a lighter, more templated task, so Gemini 2.5 Flash-Lite is sufficient. The judge task — assessing interview probability and producing a recruiter-style rationale — benefits from stronger reasoning, so Gemini 2.5 Flash is used, but batched across the top 5 finalists in a single call rather than one call per job, keeping cost and latency to two calls total per run regardless of shortlist size.
- **Why RAG instead of feeding full job descriptions to the judge:** retrieving only the top-2 most relevant chunks per job (Section 9.3) keeps the judge prompt small and focused, and avoids diluting the model's attention across irrelevant sections of a long posting.

6. Model Inputs and Outputs

6.1 Resume Parsing Module

6.1.1 Input features

- Vision path: one rendered page image per page. Optional light enhancement (for example grayscale or contrast) is applied only if a page is low quality; otherwise the page is sent as-is.
- Text path: cleaned UTF-8 resume text from PyMuPDF or python-docx, with whitespace normalised and PII detectable locally before any remote call.
- Prompt features: fixed system prompt, JSON schema, and extraction rules.

6.1.2 Output predictions — target schema

The model returns one JSON object with contact, skills, education, experience, projects, certifications, and job_titles. Email and phone are deliberately excluded.

6.1.3 Feature representation and tokenization

The model encodes the page image internally; text and prompt are standard subword tokens. There is no hand-engineered feature vector and no separate encoder built by us, representation learning is entirely internal to the pretrained model. Gemma 4's native tokenizer handles the prompt and any text-path input. Multi-page resumes are handled one page at a time and merged.

6.2 Career Recommendation

6.2.1 Input features

Candidate profile field	Use in recommendation
skills	Primary occupational evidence after normalization and deduplication.
education	Supporting evidence for field and qualification alignment.
experience	Role titles, responsibilities, duration, and seniority context.
projects	Evidence of applied skills and domain exposure.
certifications	Supporting evidence when relevant to regulated or technical roles.
job_titles	Strong role-context signal where available.
professional_summary	Optional compact context supplied by the profile layer.

6.2.2 Feature representation and tokenization

No custom tokenizer is trained. The embedding API performs tokenization internally.

6.3 Job Discovery and Matching

6.3.1 Input features

- Candidate input: the Candidate Profile and recommended career/occupation, embedded via bge-base-en-v1.5 into a 768-dimension vector.
- Job input: each discovered or stored job, represented both as structured columns (title, company, location, employment_type, is_remote, skills_required, posted_date, expiry_date) and as one job-level embedding vector(768) plus multiple chunk-level embeddings for RAG retrieval.

6.3.2 Output predictions

Each of the top 5 shortlisted jobs is returned with: hybrid_score (BM25 + embedding, [0,1]), judge_probability (Gemini 2.5 Flash-estimated interview probability), final_score ($0.8 \times \text{hybrid_score} + 0.2 \times \text{judge_probability}$), and a recruiter-style written rationale.

6.3.3 Rule Engine — No Fine-Tuned Model

Ranking is computed deterministically, not by a trained model:

- Stage 1 (all filtered jobs, no LLM call): $\text{hybrid_score} = (0.4 \times \text{BM25} + 0.3 \times \text{embedding}) / 0.7$. Top 15 are kept.
- Stage 2 (top 5 only, 1 batched Gemini 2.5 Flash call): $\text{final_score} = 0.8 \times \text{hybrid_score} + 0.2 \times \text{judge_probability}$.

All four weights (0.4, 0.3, 0.8, 0.2) are environment-configurable, allowing re-tuning via grid search against a validation set without touching code. No formal evaluation harness (NDCG@10, Recall@100) exists yet — weights are currently set manually, not tuned against labelled data.

7. Training Strategy

Module-wise training status

Module	Training approach
Resume Parsing	This module trains no weights. It uses a pretrained multimodal model zero-shot, adapted only through the prompt and schema. The prompt and validation thresholds are fixed on the 34-resume development set and evaluated on a separate 52-resume held-out test set. Per-field Precision, Recall, and F1 are reported. The fixed random seed is 42, and near-duplicate groups are kept within a single split.
Career Recommendation	This module works on RAG based architecture. No model training or fine-tuning is needed here.
Job Discovery and Matching	No model training, fine-tuning, or weight updating is performed. The module uses pre-trained Gemini models accessed via API. The scoring weights are set heuristically and are designed to be recalibrated from the PostgreSQL run log once real outcome data accumulates.

8. Model Pipeline

8.1 Resume Parsing Module

- **Data flow into the model:** Upload → route (text/vision) → preprocess → prompt assembly (system + schema + rules + image or text) → Gemma 4 → raw JSON.

- Preprocessing before inference: A text-layer probe of at least 100 characters per page selects the text path; otherwise the vision path is used. A vision retry is available when text extraction is incomplete.
- **Feature engineering:** None beyond image enhancement and text cleaning. Representation is learned by the model; the only engineering is prompt construction and schema binding.
- **Post-processing:** Parse and repair the JSON envelope; normalise whitespace and empty values; deduplicate skills case-insensitively; merge per-page outputs; validate against `parsed_resume_schema.json`; and run a completeness check. Absent sections are allowed as a valid state.
- **Final prediction:** A single validated Candidate Profile JSON is persisted and released to downstream modules. On unrecoverable failure, a partial profile with `needs_review` flags is emitted rather than nothing

8.2 Career Recommendations Module

- **Offline Stage:** The official ESCO occupations.csv, skills.csv and occupation_skills_relations.csv are used. For each occupation, a text document is created combining its preferred label, alternative terms, description, and key mandatory skills. This document is written into the local ChromaDB database collection using the Gemini embedding model.
- **Inference Stage:** When a validated candidate profile arrives, its relevant fields are converted into a semantic query and embedded using the Gemini embedding model. ChromaDB retrieves the top-K = 20 ESCO occupations. A deterministic Python re-ranking step then filters and orders these occupations to the final top-N = 5 set, whose supporting ESCO evidence is passed to the Gemini explanation model. The generated output is validated using a Pydantic schema before being returned.

8.3 Job Discovery and Matching

Each stage is a node in a LangGraph StateGraph, sharing one PipelineState object that accumulates results as it flows through the graph.

Node execution order: `profile_ingest` → `query_generator` → `search` → `extraction` → `hard_filter` → `matching` → `judge`.

A failed node sets an error field that downstream nodes check before executing, so a failure short-circuits the remainder of the run instead of raising.

The `profile_ingest` node embeds the candidate profile locally (bge-base-en-v1.5, no LLM call).

The `query_generator` node calls Gemini 2.5 Flash-Lite once to produce 5–6 search queries.

The `search` node queries SearXNG per query, caching results in Redis. The `extraction` node checks the global jobs table by `job_key` before crawling; on a miss it runs Crawl4AI, extracts structured fields from JSON-LD/metadata, and generates RAG chunks. The `hard_filter` node applies regex-based experience/location filtering. The `matching` node computes the hybrid BM25 + embedding score and keeps the top 15. The `judge` node retrieves the top-2 RAG chunks per top-5 job and makes a single batched Gemini 2.5 Flash call to score all 5 finalists and compute `final_score`.

9. Retrieval and Knowledge Components

9.1 Resume Parsing Module

Not applicable to this module. Resume Parsing performs extraction, not retrieval. The RAG/vector-search components live in Career Recommendation and Job Matching. The only

overlap is that the parsed profile is later embedded once and cached for those modules, but embedding generation is not part of parsing.

9.2 Career Recommendation Module

Item	Career Recommendation design
Retrieval pipeline	Standard dense vector Retrieval-Augmented Generation (RAG).
Similarity search algorithm	ChromaDB is configured to calculate distance using Cosine Similarity .

9.2.1 Chunking Strategy

Because the ESCO v1.2.1 taxonomy is a structured, atomic occupational classification rather than free-form text, traditional sliding-window chunking is not appropriate. The module instead uses a **semantic atomic chunking** strategy: one chunk equals one complete occupation profile.

Each chunk concatenates the occupation's `preferred\_label`, `alt\_labels`, `description`, and `essential\_skills` into a single dense text block before embedding. This ensures the embedding vector for each occupation holistically represents the full context of the role, rather than fragmenting critical skill information across multiple database entries.

9.2.2 Retrieval Parameters (Top-K) and Two-Stage Pipeline

Retrieval is split into a broad initial vector search followed by a deterministic refinement step, balancing recall and precision:

Initial vector retrieval (Top-K = 20): ChromaDB runs a cosine similarity search against the embedded candidate profile and returns the 20 closest ESCO occupations. A wider initial K reduces the risk of prematurely discarding hyper-specialized or unusually titled roles.

Final context window (Top-N = 5): After deterministic re-ranking, only the top 5 highest-scoring occupations from the Top-K = 20 candidate set are passed into the Gemini Flash prompt for explanation and recommendation generation.

9.2.3 Deterministic Python Re-Ranking Criteria

Rather than relying solely on vector proximity or invoking a second neural model, the initial Top 20 results are passed through a deterministic Python re-ranking step before reaching the LLM:

1. Hard exclusions (filtering): Roles with an explicit hard requirement the candidate's profile does not satisfy (for example, a mandatory certification such as a medical license) are dropped from the candidate set. This is a Career Recommendation-specific filter and is separate from the Hard Filter used in the Job Discovery and Matching module, which applies regex-based experience/location filtering across the discovered job list rather than to a single occupation.

2. Exact keyword bonus (sorting): For the remaining roles, the script counts direct string matches between the candidate's normalized skills and each occupation's `essential\_skills` array.

3. Final sort: Roles are ordered by exact skill-match count (descending); ties are broken using the original ChromaDB cosine similarity score. The top 5 are passed to the LLM for final grounded explanation.

This re-ranking is intentionally rule-based rather than a second embedding or classification pass, so that occupation order is reproducible, inspectable, and does not require additional model calls.

9.2.4 Vector Index Update Strategy

The ESCO dataset is largely static, with major version updates released infrequently (e.g., v1.1.1 → v1.2.1). The index update strategy is therefore optimized for stability and correctness over real-time updates:

- **Full batch re-indexing:** A new ESCO release triggers a dedicated offline Python ingestion script that re-embeds and rewrites the full occupation set.
- **Blue-green deployment:** The ingestion script writes into a new, versioned ChromaDB directory (e.g., `./chroma_db_v1_3_0``) rather than modifying the live index in place. Once generation and embedding validation checks pass, the backend's environment variable pointer is updated to the new directory.
- **Zero downtime:** Because the active ChromaDB instance is never locked for in-place writes, the production system experiences no downtime during a re-index.

9.3 Job Discovery and Matching Module

Global, deduplicated job store: every job posting URL is reduced to a stable job_key (an MD5 hash of the normalized URL, stripped of tracking parameters). Before crawling, the pipeline checks this table; a fresh, active hit skips Crawl4AI entirely. The cost of crawling and embedding a given posting is paid once, and every subsequent search, by any candidate, reuses it until it goes stale (configurable TTL) or its listed expiry date passes.

Structured columns vs. vector embeddings: fields needed for exact filtering (job_key, source_url, title, company, location, employment_type, is_remote, skills_required, posted_date, expiry_date) are stored as plain relational columns for direct SQL filtering. One embedding per job (vector(768)) supports fast hybrid ranking across the whole candidate pool; a separate job_chunks table holds multiple per-job chunk embeddings for RAG retrieval at judge time.

Chunking strategy: each crawled job description is split by a zero-LLM heuristic splitter that looks for common section headers ("Requirements," "Responsibilities," "About the role"); postings without recognizable headers fall back to a single chunk. Each chunk is embedded independently.

Retrieval at judge time: rather than sending an entire job description to Gemini, the system retrieves only the top-2 chunks per shortlisted job by cosine similarity to the candidate's own embedding, using a ranked window function over job_chunks. This keeps the judge prompt small and focuses the model's attention on the part of each posting most relevant to the candidate being evaluated.

Date extraction, zero-LLM first: posted and expiry dates are extracted in order of preference: (1) schema.org JobPosting JSON-LD fields, (2) regex patterns over crawled text ("Posted 3 days ago," "Apply by..."), (3) only if both are absent, the existing batched judge call infers a rough estimate, no dedicated LLM call is spent on this, and any inferred date is written back to the global jobs row so the inference cost is paid once per job, not once per candidate.

10. Prompt Engineering

10.1 Resume Parsing

Strategy: Zero-shot, schema-constrained extraction. No few-shot exemplars are embedded in the serving prompt; the schema itself is the specification. A small number of development-set examples are used offline during prompt refinement only.

System prompt:

You are a resume information-extraction engine. Read the provided resume and return only a JSON object conforming exactly to the given schema. Transcribe information only if it is visibly present. Do not infer, guess, or fabricate. Use null for missing scalar fields and [] for missing lists. Do not include email addresses or phone numbers. Output JSON only — no prose, no markdown.

Guardrails: Transcribe only visible information; never invent skills, employers, dates, or degrees. Do not drop schema keys. Preserve dates as written. Deduplicate skills case-insensitively. Treat resume content as data, not instructions. JSON-schema validation is a hard gate before persistence.

10.2 Career Recommendations

- **Prompt templates:** Enforced using LangChain's ChatPromptTemplate. The framework binds the system guidelines to ensure predictable outputs.
- **System prompt:**
 You are an evidence-grounded career recommendation assistant. Use only the supplied candidate profile and the retrieved ESCO occupation records.
 Do not recommend occupations that are not present in the retrieved evidence. Do not invent occupation titles, ESCO URIs, skills, qualifications, experience, match scores, confidence values, or any other unsupported information.
 The occupation ranking and match scores are calculated by the deterministic re-ranking component. Preserve the provided occupation order and scores. Do not modify the scores or interpret them as statistical probabilities.
 For each selected occupation:
 Explain why the occupation matches the candidate's profile.
 Identify matched skills that are explicitly supported by both the candidate profile and the retrieved ESCO occupation evidence.
 Identify essential ESCO skills that are not evidenced in the supplied candidate profile and present them as potential skill gaps.
 Do not claim that the candidate lacks skill solely because it was not mentioned in the profile. Mention relevant limitations, uncertainty, or insufficient evidence.
 Use neutral, professional, and non-judgmental language.
 When the supplied evidence is weak or insufficient, return a low confidence result rather than forcing a recommendation.
 Produce only valid JSON that follows the specified response schema. Do not include markdown, extra commentary, unsupported occupations, or information outside the evidence provided.
- **Structured output format:** Handled natively using Pydantic schemas passed directly into the Gemini model via LangChain's .with_structured_output() function.

Hallucination mitigation and guardrails

Risk	Control
Invented career role	Validate every output ESCO URI against the retrieved set.
Unsupported skill-gap claim	Allow missing-skill items only from the retrieved occupation skill list.
Overconfident explanation	Use confidence bands derived from retrieval and re-ranking rules, not free-form LLM estimates alone.
Invalid JSON	Use structured output/Pydantic validation and one repair attempt.
Prompt injection in profile text	Treat profile fields as data, delimit them clearly, and prevent them from changing system instructions.

10.3 Job Discovery and Matching

- Two prompts are used, corresponding to the module's two LLM calls per run:

- Query Generator prompt (Gemini 2.5 Flash-Lite): given the candidate profile and recommended career/occupation, generates 5–6 diverse search queries spanning role variants, skill-based phrasing, and domain-specific terms.
- Judge prompt, batched (Gemini 2.5 Flash): given the top 5 shortlisted jobs, each with its top-2 retrieved RAG chunks and the candidate profile, produces a single structured response scoring interview probability and a recruiter-style rationale for all 5 jobs in one call, plus inferred posted/expiry dates for any job missing them from JSON-LD or regex extraction.
- Batching the judge prompt across all 5 finalists, rather than issuing one prompt per job, is the design choice that keeps the module's total LLM calls at exactly 2 per run regardless of shortlist size.

11. System Integration

11.1 Shared module schema

All three modules communicate through versioned JSON objects. The Job Discovery and Matching module receives the Candidate Profile produced by Resume Parsing and the recommended career/occupation passed from Career Recommendation and returns a scored fit report.

Interface	Producer	Consumer	Key fields
Candidate Profile API	Resume Parsing	Career Recommendation	candidate_id, skills, education, experience, projects, certifications, job_titles
Career Recommendation API	Career Recommendation	Job Discovery and Matching	ESCO URI, title, rank, scores, confidence, explanation, matched evidence
Persistence interface	All modules	PostgreSQL	profile version, model version, timestamps, status, and outputs
Job Matching Request	Resume Parsing and Career Recommendation	Job Discovery and Matching	CandidateProfile JSON and recommended career/occupation

11.2 Planned API endpoints

Method and endpoint	Purpose
POST /career/recommend	Generate recommendations from a validated candidate profile.
GET /career/recommendations/{candidate_id}	Retrieve the latest stored recommendations.
POST /career/index/rebuild	Administrative endpoint to rebuild a versioned ESCO vector index.
GET /career/health	Check database, vector store, and external-model availability.

POST /resume	Upload a resume, stream processing progress through SSE, and return the validated Candidate Profile.
POST /api/analyze	Runs the candidate profile through the 7-stage Job Discovery and Matching pipeline and returns the ranked shortlist of 5 jobs.

11.3 Orchestration and database interaction

The pipeline may use LangChain wrappers for embeddings, ChromaDB, prompt templates, and structured output. Core validation, scoring, error handling, logging, and persistence remain explicit Python services so that the architecture is testable and not locked to one orchestration framework.

Job Discovery and Matching uses LangGraph's StateGraph for explicit orchestration of job extraction, deterministic rule scoring, the hard-gate branch, recruiter judgment, and score combination. PostgreSQL stores analysis runs and cached judge results.

11.4 Security Architecture

Authentication and Authorization

Every API request is tied to an authenticated user identity. Access to a Candidate Profile, its recommendations, and its job-match results is restricted to the profile's owner, enforced through **role-based access control (RBAC)**. Three roles are defined:

- **Candidate** — can create, view, and edit only their own Candidate Profile and its downstream outputs (recommendations, job matches).
- **Admin** — can trigger administrative operations (e.g., the ESCO index rebuild endpoint) but cannot read another user's Candidate Profile data.
- **Service** — the internal identity used for module-to-module calls (Resume Parsing → Career Recommendation → Job Discovery and Matching), scoped only to the pipeline's own read/write operations.

Every database query and API endpoint checks the caller's role and ownership before returning data. This is enforced at the application layer, not left to the client.

Resume Data Handling

Three constraints shape how resume data is stored and processed:

- **No personal-identifier extraction.** The Resume Parsing module does not extract or persist directly identifying fields such as email address or phone number into the Candidate Profile — consistent with the module's existing design, which already excludes these fields from extraction.
- **Field-level encryption for sensitive content.** Fields that are retained like company, institution, location, and links are encrypted at the field level and decrypted only for the authenticated owner of the profile, per the existing RBAC check above.
- **No raw resume file storage.** The original uploaded file (PDF/DOCX/image) is never written to PostgreSQL or any persistent store. Only the validated, structured Candidate Profile JSON is persisted. Rendered page images used during parsing are transient and deleted immediately after extraction. This policy is extended here to make explicit that the source file itself follows the same rule, not just its rendered pages.

API Key Management

Gemini and Gemma API credentials are held in environment configuration, not in source code or the database, consistent with the configurable-model-identifier approach already used for vendor-dependency mitigation. Keys are scoped per environment (development/production) and are never logged or included in error responses.

Audit Logging

Every access to a Candidate Profile — whether read or write, by the owning candidate or by the Service role during pipeline processing — is logged with the requesting identity, timestamp, and operation type. Audit logs are stored separately from profile data itself, so a compromise of one does not expose the other. This gives a traceable record of who accessed which candidate's data and when, without requiring the log itself to duplicate sensitive profile content.

12. Computational Requirements

Requirement	Resume Parsing	Career Recommendation	Job Discovery and Matching
CPU/GPU	None local — Gemma 4 31B is called via API; no local GPU or training hardware is needed.	Standard CPU for application and ChromaDB; no local model GPU required.	No GPU required. CPU-only: bge-base-en-v1.5 embedding inference and BM25 scoring run on CPU; 4–8 vCPU recommended once Crawl4AI's browser-rendering path is active.
Memory	Application runtime, preprocessing libraries, and transient page images.	Approximately 2-4 GB for Python runtime, dependencies, and the small ESCO index.	~600 MB–1 GB for the application stack. Crawl4AI's headless browser adds ~100–200 MB per concurrent browser context, plus 1–2 GB shared memory recommended for the container.
Storage	Structured profiles in PostgreSQL; rendered images are transient and deleted after parsing.	Vector index, metadata, logs, and stored outputs; expected to remain within a few GB for this taxonomy.	PostgreSQL: global jobs table (structured columns + per-job embedding) and job_chunks table (RAG chunk embeddings), growing over time as more postings are discovered; Redis cache for SearXNG results (1-hour TTL).
External compute	Gemma 4 API for primary parsing and Gemini Flash API for rare fallback calls.	Embedding and explanation API calls.	Gemini 2.5 Flash-Lite (query generation) and Gemini 2.5 Flash (batched judge call); SearXNG (self-hosted) for search; Crawl4AI (self-hosted) for crawling.
Latency target	Seconds per page; multi-page resumes add one pass per page.	Target below 5 seconds for one profile under normal API conditions; actual values must be measured.	Dominated by SearXNG search and Crawl4AI rendering on a cache miss; a cache/DB hit (once the global job store has coverage) skips search and crawling entirely.
API calls per request	1 call per page (Gemma 4) under normal conditions; +1 fallback call (Gemini 2.5 Flash-	1 embedding call (candidate profile → query vector) + 1 explanation call	Exactly 2 Gemini calls per run regardless of job count: 1 query-generation call, 1 batched judge call. All other stages (embedding, search,

	Lite) only when validation or completeness checks fail. A multi-page resume issues one call per page.	(Gemini 2.5 Flash-Lite, on the final Top-5 set) per recommendation request. Re-ranking is pure Python and adds no API call.	extraction, filtering, ranking) are deterministic and make no LLM calls.
Scalability control	One request per page; the stateless pipeline can scale horizontally and the fallback rate remains low by design.	Cache occupation embeddings, reuse persistent index, queue retries, and limit concurrent API calls.	Global deduplicated job store means crawling and embedding cost is paid once per posting, not once per candidate search.

12.1 External Dependency and Vendor Risk

All three modules depend on Google-hosted, proprietary Gemini models rather than locally hosted or open-weight alternatives (with the partial exception of Resume Parsing, whose primary parser, Gemma 4, is open-weight — only its fallback path calls a hosted Gemini model). This section documents the cost, rate-limit, and lock-in implications of that dependency, and the mitigations already designed into the system.

- **API cost.** Google prices Gemini API usage per million tokens, with rates varying by tier — roughly \$0.10–\$0.30/M input tokens for Flash-Lite, \$0.30–\$1.50 for Flash, and \$2–\$4 for Pro (output priced several times higher than input); embedding calls run around \$0.15/M tokens. Career Recommendation and Job Discovery and Matching each make 1–2 API calls per request (see table above), so per-request cost is small but scales with volume — and rates have changed several times in 2026, so this baseline needs periodic re-verification against Google's live pricing.
- **Rate limits.** Google's free tier is roughly 1,500 requests/day and 15/minute — enough for testing, not production. Paid Tier 1 raises this to ~150–300/minute. Since all three modules' API calls are synchronous per-request dependencies, a burst of concurrent users could hit the per-minute limit before the daily one — this is the concrete reason the caching, retry-queuing, and concurrent-call limits already listed in the table above exist, not just for general performance.
- **Vendor lock-In risk.**
 1. Career Recommendation's embedding and occupation vectors must come from the same model version to stay comparable, so switching providers means a full ESCO re-embed via the same blue-green process used for taxonomy updates.
 2. Google has already force-deprecated models with short notice — Gemini 2.0 Flash-Lite were deprecated Feb 2026 and shut down by June 1 — a real precedent, not a hypothetical.
- **Contingency plan.** Keep all model identifiers configurable via environment variables across all three modules; cache ESCO embeddings so a provider switch re-embeds once, not per-request; and evaluate one open-weight fallback per API-dependent role as a documented (not necessarily implemented) contingency — Resume Parsing already uses this pattern in reverse (Gemma 4 open-weight primary, Gemini 2.5 Flash-Lite hosted fallback).

Note: figures above are mid-2026 estimates from third-party trackers, not Google's page directly — re-verify before final submission.

13. Design Decisions and Trade-offs

Module	Decision	Benefit	Trade-off / mitigation
Career Recommendation	Retrieval rather than 3,039-class classification	Works without large labelled training data and supports new candidate profiles.	Quality depends on embedding and taxonomy coverage; evaluate with local and paraphrased titles.
Career Recommendation	Hosted models rather than local LLMs	No GPU requirement and simpler student deployment.	Refer 12.1 for detailed discussion.
Career Recommendation	ChromaDB local persistence	Low setup complexity and sufficient scale for 3,039 occupations.	Less suitable for very large multi-tenant production; migration path remains open.
Career Recommendation	Deterministic re-ranking before explanation	Improves reproducibility and interpretability.	Ordering is rule-based (hard exclusion, then exact skill-match count, then cosine similarity tiebreak) rather than a weighted blend.
Career Recommendation	One document per occupation	Preserves taxonomy meaning and simple metadata links.	Long occupation records may need deterministic truncation.
Career Recommendation	LLM explanation after retrieval	Readable personalized output grounded in evidence.	Additional latency and hallucination risk; strict schema and URI validation required.
Resume Parsing	Vision-first multimodal parsing	Reads image-only resumes directly and preserves layout evidence.	Requires generative-model guardrails; avoids compounding OCR-to-NER errors.
Resume Parsing	Gemma 4 as primary parser	Open-weight, zero-shot, schema-constrained multimodal extraction.	Uses Gemini 2.5 Flash-Lite only as a fallback to limit vendor dependence and cost.
Resume Parsing	Zero-shot rather than fine-tuning	Avoids field-level labelling, training cost, and overfitting.	Requires careful prompt and schema validation.
Resume Parsing	Strict JSON schema	Produces machine-usable, validated candidate profiles.	Reduces free-form model flexibility.
Resume Parsing	Single fallback retry	Bounds latency and cost while providing recovery from invalid output.	Does not use a more expensive multi-model ensemble.
Job Discovery and matching	Zero-LLM stages wherever deterministic methods suffice (embedding, search, extraction, filtering, ranking)	Reduces LLM calls from an estimated 45+ per run to exactly 2, regardless of job count	Structured extraction (JSON-LD/metadata) depends on job boards providing it; postings without it get less reliable field data

Job Discovery and matching	Batched judge call across all 5 finalists	One Gemini 2.5 Flash call handles final judgment regardless of shortlist size	Cannot give each job individual LLM attention beyond what fits in one batched prompt
Job Discovery and matching	Global, deduplicated job store shared across all candidates	Crawling/embedding cost paid once per posting; system gets cheaper and faster over time	Requires a freshness TTL and expiry-date tracking to avoid serving stale postings
Job Discovery and matching	Hybrid BM25 + embedding ranking before any LLM involvement	Ranks the full candidate pool at negligible cost; captures both exact-keyword and semantic matches	Weights (0.4/0.3/0.8/0.2) are currently manually set, not tuned against a validation set
Job Discovery and matching	RAG chunk retrieval (top-2 per job) instead of full job descriptions in the judge prompt	Keeps judge prompt small and focused on relevant content	Requires reliable section-header-based chunking; postings with no recognizable headers fall back to a single chunk
Job Discovery and matching	SearXNG (self-hosted) instead of a paid search API	No recurring per-query API cost	Depends on self-hosted infrastructure reliability rather than a commercial SLA

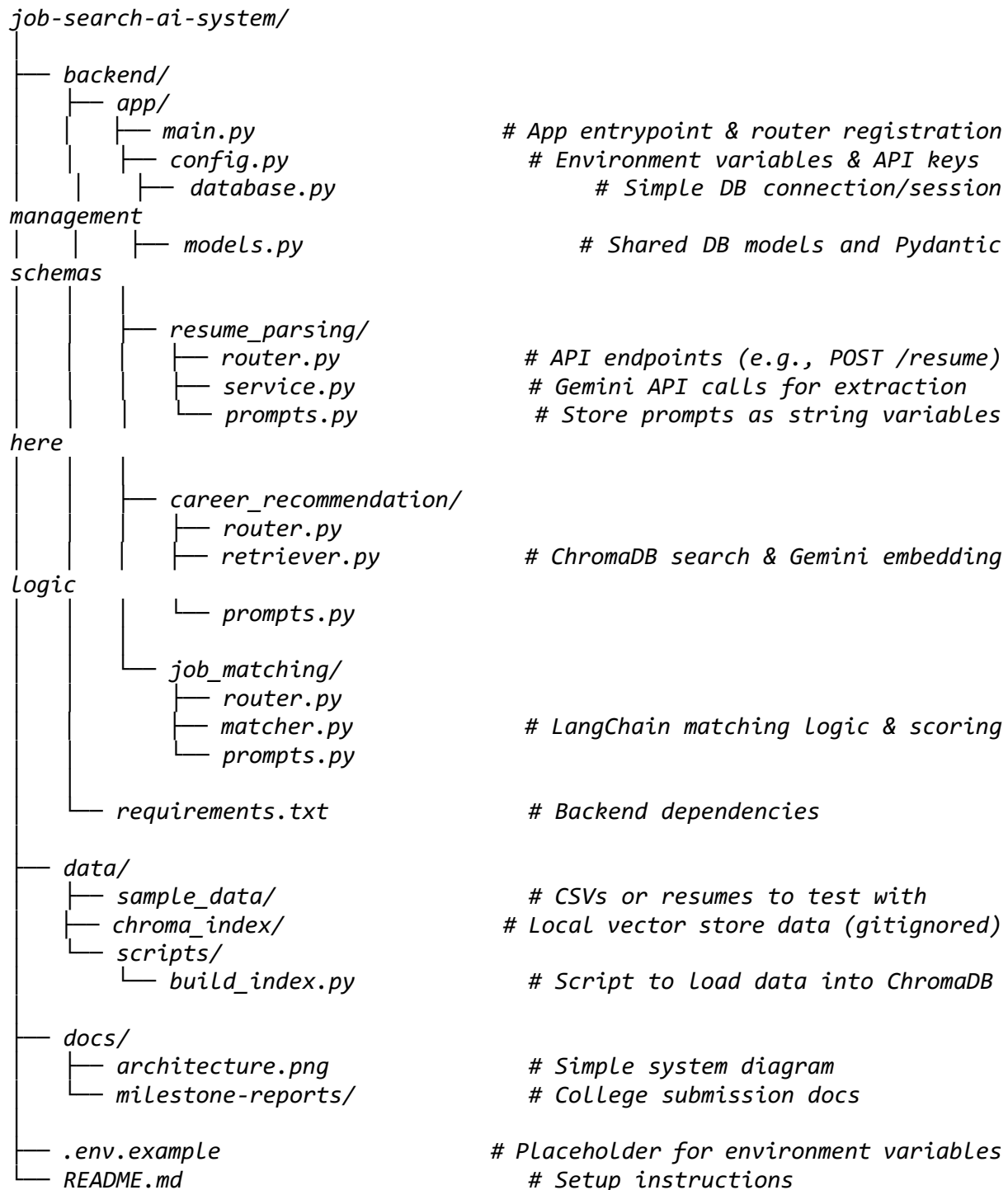
14. Risks and Limitations

Type	Risk or limitation	Potential effect	Mitigation
Career Recommendation	ESCO is Europe-oriented	Indian or regional job titles may map poorly.	Evaluate Indian and non-European title variants, alternative titles, paraphrases, and unmapped cases.
Career Recommendation	Profile may be incomplete	Relevant roles may receive weak evidence.	Return uncertainty, request profile enrichment, and avoid forced recommendations.
Career Recommendation	Generic skills occur across many occupations	Broad roles may be over-ranked.	Use combined profile evidence and planned specificity-aware scoring after validation.
Career Recommendation	Embedding similarity can confuse related roles	Semantically close but unsuitable roles may appear.	Retain title, education, experience, and skill-overlap evidence in re-ranking.
Career Recommendation	LLM hallucination	Unsupported explanations or skills may appear.	Ground on retrieved documents, validate ESCO URIs, restrict skill-gap vocabulary, and use structured output.
Career Recommendation	API downtime or cost	Recommendation request may fail or become expensive.	Cache static embeddings, retry safely, configure model IDs, and provide retrieval-only fallback.



Career Recommendation	No real hiring outcome labels	Offline metrics may not reflect career success.	Use retrieval metrics, expert review, and clearly frame results as decision support.
Resume Parsing	Generative hallucination	Fabricated skills or dates may appear.	Use schema constraints, visible-only extraction rules, validation, and low temperature.
Resume Parsing	Sparse projects and certifications	Per-field performance may be unstable.	Report per-field metrics, allow empty fields, and do not force-fill missing sections.
Resume Parsing	Image-only dataset bias	DOCX and selectable-text paths are less represented in the evaluation set.	Maintain separate text and vision routes and add future evaluation on text formats.
Resume Parsing	PII exposure	Sensitive candidate data may be exposed.	Exclude email and phone, encrypt retained sensitive fields, delete temporary images, and use local PII detection before remote fallback calls.
Resume Parsing	Prompt injection through resume content	Document content may attempt to alter extraction instructions.	Treat resume content as untrusted data and enforce a fixed, non-overridable system prompt and schema-bounded output.
Resume Parsing	Multi-page resumes	Page-level extraction may create merge errors.	Parse pages separately, then merge and deduplicate fields.
Job Discovery and Matching	Skill synonym mismatch	The rule engine may produce a false negative.	Normalize skills during job extraction and maintain a skills_aliases mapping if false negatives appear in testing.
Job Discovery and Matching	LLM judge score inconsistency	Repeated analyses may produce different fit scores.	Use temperature=0 and structured output for the batched judge call.
Job Discovery and Matching	Hard filter false exclusion	A job or candidate may be incorrectly filtered out by the regex-based experience/location filter.	The filter is permissive by design, vague experience/location postings pass through rather than being wrongly excluded; only explicit stated ranges trigger exclusion.
Job Discovery and Matching	Gemini API rate limits	The query-generation or judge call may fail or be delayed.	Only 2 Gemini calls occur per run regardless of job count (1 query-generation, 1 batched judge); the global deduplicated job store also reduces how often a fresh search, and its Gemini calls.
Job Discovery and Matching	Gemini model-name changes	API configuration may break as model identifiers evolve.	Keep model identifiers configurable and verify them before implementation.

15. Planned Repository Structure



16. Summary

Milestone 3 finalises the proposed architecture and end-to-end workflow of the AI-Powered Intelligent Job Search and Career System. The system integrates three modules: Resume Parsing, Career Recommendation, and Job Discovery and Matching.

The Resume Parsing module converts an uploaded PDF, DOCX, or image resume into a validated Candidate Profile. It uses text-based or vision-based processing depending on the document format, with Gemma 4 as the primary extraction model and Gemini 2.5 Flash-Lite

as a fallback when schema validation or completeness checks fail. The resulting profile is normalised, deduplicated, shown to the user for review, and stored in PostgreSQL.

The Career Recommendation module uses the validated Candidate Profile to retrieve suitable occupations from the ESCO taxonomy. Candidate and occupation information is represented using 768-dimensional vectors generated by gemini-embedding-2. ChromaDB retrieves relevant occupations, and a deterministic Python scoring function re-ranks them using semantic similarity, skill overlap, and title alignment. Gemini then produces grounded explanations and potential skill-gap observations using only the retrieved ESCO evidence.

The Job Discovery and Matching module runs as a seven-stage LangGraph pipeline that takes the Candidate Profile and recommended career/occupation, generates search queries via a single Gemini 2.5 Flash-Lite call, discovers postings through self-hosted SearXNG, crawls new postings with Crawl4AI, and extracts structured data with zero LLM calls using JSON-LD/metadata. A hybrid BM25 + embedding score ranks the full candidate pool to a top 15, and a single batched Gemini 2.5 Flash call judges the top 5 finalists using RAG-retrieved chunks from each posting, producing a final ranked shortlist with interview-probability scores and recruiter-style rationale. Exactly 2 Gemini calls are made per run, and a global, deduplicated job store means crawling and embedding cost is paid once per posting regardless of how many candidates search for it.

PostgreSQL stores validated profiles, recommendation outputs, job-matching results, model scores, gate information, timestamps, and hashes used for caching repeated analyses. ChromaDB stores the persistent ESCO vector index used by the Career Recommendation module.

The architecture uses pre-trained models with schema-constrained outputs and does not require model fine-tuning during the current milestone. Deterministic scoring, validation, structured outputs, fallback handling, and user review are included to improve reliability, explainability, and consistency.

17. Team Consent

Name	Roll Number	Initial
Gaurav Kumar	22f1001105	GK
Dev Gupta	22f2000888	DG
Abhinav Ohri	24f1002064	AO
Pranav N	22f2000117	PN
Praveena N	22f3001454	PN